Lecture 6

Compiler Design CSC-305

Awanish Pandey

Department of Computer Science and Engineering
Indian Institute of Technology
Roorkee

August 3, 2026

How to reduce development and testing effort?

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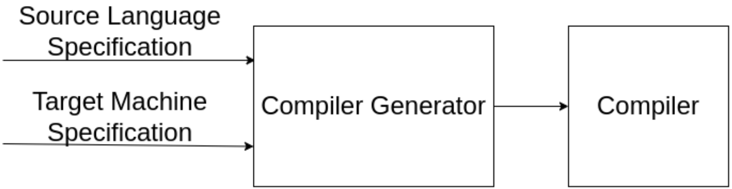

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- Cuts down development/maintenance time by almost 30-40%
- Tool development and testing is a one-time effort
- Compiler performance can be improved by improving a tool

Lexical Analysis

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tokens	id	ASSIGN	id	ADDOP	ID
lexeme	a	$=$	b	$+$	c

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- Use high-level languages like C
  Efficient but difficult to implement
- Use tools like `lex`, `flex`
  Easy to implement but not as efficient as the first two cases

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- The first character cannot determine what kind of token we are going to read

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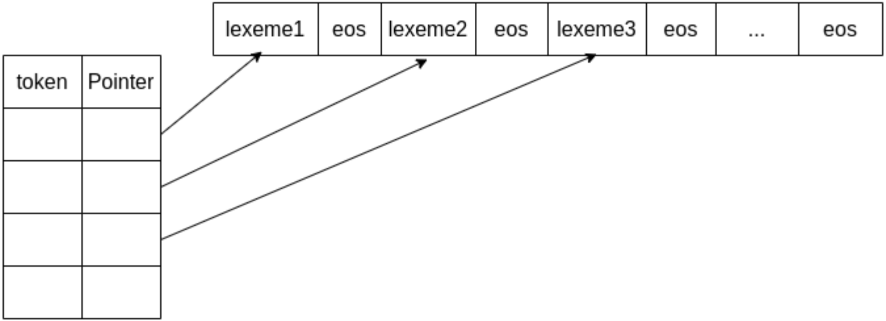

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- Even today `Foo<Bar<Bazz>>`

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  - Each character should be looked at only once.

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- Where each r_i is a regular expression over $\sum \cup d_1 \cup d_2 \cup \ldots \cup d_{i-1}$